

PSI Exam Prep: Real Estate Calculations

STUDY SHEET

Top Takeaways

Seven percent of your exam questions will relate to financing and real estate calculations.

A. Calculations for Transactions

1. Seller's Net Proceeds

- The net-to-seller formula calculates the amount the seller receives after paying commission and other expenses.
- The closing officer ensures that the seller's loan and all interest is paid unless the buyer is assuming the loan. The seller pays per diem interest for the month of closing.
 - Percent to seller = 100% of sale price - commission %
 - Net to seller = sales price x percent to seller

Example

Sales price is \$150,000 with a 7% commission rate. Percent to seller is 93% or 0.93. Net to the seller = $\$150,000 \times .93 = \$139,500$.

- Number of days in the month of closing $\times$ per diem interest = seller interest owed at closing

Example

Using a statutory year, the seller's loan balance is \$60,000 with a 5% interest rate.

The sale closes on June 15, with the last payment made on June 1.

Seller interest owed = $(\$60,000 \times 5\%) \div 360 = \$8.33 \times 15 = \$124.95$

2. Buyer Funds Needed at Closing

- At closing, buyers are expected to have funds available to meet the seller's purchase price. A portion of these funds will have already been paid in the form of the earnest money (good faith) deposit. The buyer must bring the remainder to closing in either personal or lender funds.
- Buyers owe closing costs, lender fees, a portion of the upcoming year's property taxes and insurance, recording fees, etc.
- Buyers with a new loan also usually owe interest at closing because the first monthly payment is generally made on the first day of the second month after closing. The interest is based on a per diem. So, if closing is on June 15, the first monthly payment is made on August 1, and interest is calculated during all of July. Since closing was June 15, the purchaser will owe interest on this loan from the day of closing through the last day of the month.

Example

Using a statutory year, the buyer's loan is \$315,000 with a 4% interest rate.

The sale closes on June 15.

Buyer interest owed = $(\$315,000 \times 4\%) \div 360 = \$35 \times 16 = \$560$

- The earnest money deposit given in good faith with the contract is not the down payment. Earnest money is applied to the property purchase price.
- If earnest money is less than the down payment, the purchaser brings the remainder of the down payment to closing.
- If earnest money exceeds the down payment plus closing costs, the overage is refunded to the buyer at closing.
- Any funds borrowed for a down payment are included in the debt-to-income calculation.
 - Loan balance $\times$ interest rate = amount of annual interest
 - Amount of annual interest $\div$ number of days in a year (360 in a statutory year; 365 in a calendar year) = per diem interest
 - Total amount paid on a loan = monthly payment $\times$ total number of payments
 - Total amount of interest on a loan = (monthly payment $\times$ total payments) - original loan value
 - Total number of loan payments = 12 (months per year) $\times$ number of years of the loan
- Most simply, a buyer's 'cash to close' is the down payment and closing costs, minus earnest money and other credits.

3. Real Property Tax and Other Prorations

- Property taxes are the largest funding source for local services such as schools, roads, and police and fire protection.
- Calculating property taxes requires knowing the appraised value, tax assessment ratio (may vary by state), and tax rate.
 - $\text{Value} \times \text{assessment ratio} = \text{assessed value}$
 - $\text{Assessed value} \times \text{tax rate} = \text{annual property tax}$. Divide by 12 (months in a year) to get the monthly property tax amount.
- Tax rates are expressed as a rate per dollar amount of value, such as \$3 per \$100. Divide \$3 by \$100 to calculate the tax rate. So, $\$3 \div \$100 = .03$.
- Tax rates can also be expressed as mills. When mills are used, it's a rate per \$1,000. So, 30 mills is the same as $\$30 \div \$1,000$.

Example

A property worth \$350,000 is assessed at 40%.

The tax rate is \$30 per \$1,000, or 30 mills.

Annual taxes = $\$350,000 \times 40\% = \$140,000$. $\times .03 = \$4,200$

- A **proration** is calculating what "pro-rata" amount is owed by (or to) each party at closing for expenses or income generated by the real estate sale. Prorated items are either accrued or prepaid.
- **Accrued** items are amounts allocated to both buyer and seller that haven't yet been paid but will be paid by the buyer after closing. These items are credited to the buyer and debited to the seller.
- **Prepaid** items are amounts allocated to both buyer and seller that were paid by the seller before closing. These items are credited to the seller and debited to the buyer.
- You must know whether to use an actual/calendar year (365 days; 366 in a leap year) or a statutory/banker's year (360 days), how many days each party is responsible for, and the per diem rate. You need to know if the billing is accrued or prepaid and what time period is covered.
 - $\text{Per diem rate} = \text{amount to be prorated} \div \text{either 360 or 365}$
- Unless told otherwise, calculate prorations with the seller responsible for expenses and income on closing day. The seller pays expenses through closing day, and all rents due from tenants are credited to the seller through closing day.
- Mortgage interest and insurance are only prorated between buyer and seller if the buyer is assuming the seller's loan.
- Taxes and rents are generally prorated regardless of financing method.
- Prorate rent in an occupied unit to pay seller rent through closing day and pay buyer rent for the day after closing through the month's end.
 - $\text{Daily rent} = \text{monthly rent} \div \text{days in month}$
 - $\text{Prorated rent} = \text{daily rent} \times \text{number of days lessee occupied the unit}$
- If using a statutory or calendar year isn't specified, calculate rents based on actual days.
- If instructed to allocate closing day to the buyer, calculate the seller share up to closing day.

4. Real Property Transfer Fees

- **Transfer tax** is applied when real property is sold. Rates vary by state and perhaps even locality.
- Payment may be negotiable between the parties. It is typically paid on the sale price.
- A **mortgage / deed of trust recording tax** is a tax on the recordation of the mortgage/deed of trust and is based on the loan amount.
- Both taxes are usually quoted as an amount per \$100 or an amount per \$1,000, and the method for calculating is similar to taxes.
 - $\text{Amount per } \$100 = (\text{sale price} \div \text{loan amount} \div 100) \times \text{tax rate}$
 - $\text{Amount per } \$1,000 = (\text{sale price} \div \text{loan amount} \div \$1,000) \times \text{tax rate}$

5. Principal, Interest, Taxes, and Insurance (PITI) Payments Estimate Given Loan Rate and Term

- **Amortized loans** are loans that are paid off over time.
- Each periodic payment (usually monthly) is credited first toward current monthly interest on the loan, with the remainder used to pay off a portion of the principal.
- Payments on a **fully amortized loan** are sufficient to pay off the loan by the end of the loan term.
- Loan payments can be determined from an amortization chart. To use an amortization chart to calculate a monthly principal and interest payment, find the point in the amortization chart where the interest rate and the term of the loan intersect. This location yields the factor (a number that will appear as a decimal, such as 6.43509) you'll use to compute the payment. This factor is per \$1,000 of the mortgage.

Example

A 2.000 (rate %) for a 15-year loan term = a factor of 6.43509.

Monthly payment = $(\text{loan amount} \div 1,000) \times \text{factor}$

- Most mortgage payments include principal, interest, taxes, and insurance (PITI).
- Interest is calculated on the outstanding principal (loan) balance. The principal portion of each monthly payment reduces the outstanding principal balance, so the amount of interest due with each payment decreases each month.

- Lenders establish an escrow account into which 1/12 of the anticipated property taxes and homeowners insurance is paid each month. The lender then pays these bills when they come due, protecting the lender's interests in the property.
- You may be asked to perform a number of interest and payment calculations on your exam:
 - $\text{Loan balance} \times \text{interest rate} = \text{annual interest}$
 - $\text{Annual interest} \div \text{loan balance} = \text{interest rate}$
 - $\text{Annual payment} \div \text{loan balance} = \text{interest rate (convert to a \%)}$
 - $\text{Annual interest} \div \text{interest rate} = \text{loan balance}$
 - $\text{Divide annual interest by 12} = \text{monthly interest}$
 - $\text{Monthly payment} \times \text{total number payments} - \text{original loan value} = \text{total loan interest amount}$
- The monthly payment credited toward principal = interest portion - interest from the payment. The remainder goes toward principal.

Example

The loan amount is \$315,000 with a 5% interest rate and a 30-year term.

The monthly principal and interest payment is \$1,690.99.

To calculate the amount owed after the first payment is made:

1. $\text{Loan balance} \times \text{interest rate} = \text{annual interest amount } (\$315,000 \times 5\% = \$15,750)$
2. $\$15,750 \div 12 \text{ (months)} = \$1,312.50 \text{ (monthly interest)}$
3. $\text{Monthly principal and interest} - \text{first month's interest} = \text{amount applied to principal } (\$1,690.99 - \$1,312.50 = \$378.49)$

- $\text{Loan amount} - \text{amount applied to principal} = \text{remaining loan balance } (\$315,000 - \$378.49 = \$314,621.)$

B. General Concepts

1. Basic Math Tips

- When multiplying or dividing, it may be easier to first convert percentages to decimals: Divide the percentage by 100 and drop the percent sign (or move the decimal point left two places)
 - $5\% = .05$
- Convert a decimal to a percentage: Multiply by 100 and add a percent sign (or move the decimal point right two places)
 - $.045 = 4.5\%$
- To convert a fraction to a decimal, divide the top number by the bottom number.

2. PSI Exam Math Calculations

- **This information is NOT available at the test center; you should memorize it.**
- Candidates may use the following information to make mathematical calculations on the general section of the real estate examinations unless otherwise stated in the text of the question:
 - There are 43,560 square feet per acre.
 - There are 5,280 feet in a mile.
 - Round off calculations (where applicable).
- If a question requires the calculation of prorated amounts, the question will specify:
 - Whether the calculation should be made on the basis of 360 or 365 days a year.
 - Whether the day of closing belongs to the buyer or seller.

3. Net Operating Income

- To determine net operating income (NOI), begin with potential gross income (i.e., income if building were fully rented). Subtract the probable loss due to projected vacancy from potential gross income to estimate effective gross income. Subtract operating expenses from effective gross income to arrive at net operating income.
- Determine net operating income using the cap rate formula: multiply cap rate by value ($R \times V = I$)

5. Equity

- Equity is the amount of value or interest a homeowner has in a property less debts against the property (such as a mortgage loan).
 - $\text{Equity} = \text{value} - \text{amount owed}$.
- Appreciation in value increases equity.
- Depreciation in value reduces equity.

Example

Property value = \$350,000

Loan amount = \$315,000

$$\text{Equity} = \$350,000 - \$315,000 = \$35,000$$

6. Capitalization Rate

- Capitalization rate or cap rate is an annual rate of return from an income-producing property.
- Used by appraisers to estimate value of rent generating properties. Often used by investors to compare one investment to another.
 - $\text{Cap rate} = \text{NOI} \div \text{value (or } I \div V = R)$
 - $\text{Value} = \text{NOI} \div \text{cap rate (or } I \div R = V)$

8. Commission and Commission Splits

- $\text{Total commission} = \text{sales price} \times \text{commission rate}$

Example

Property sells for \$350,000 with a 5% commission.

$$\text{Commission} = \$350,000 \times .05 \text{ (or } 5\%) = \$17,500$$

- $\text{Sale price} = \text{total commission} \div \text{commission rate}$

Example

A broker earns a 5% commission of \$17,500: $\text{Sale price} = \$17,500 \div 5\% = \$350,000$

- $\text{Commission rate} = \text{total commission} \div \text{sale price}$

Example

A broker earns \$17,500 for selling a \$350,000 property.

$$\text{Commission rate} = \$17,500 \div \$350,000 = .05 \text{ or } 5\%$$

9. Loan-to-Value Ratio

- The loan-to-value (LTV or LTVR) ratio is the percentage of a property's value the bank loaned/will loan on. It's the ratio of the loan amount to the property's value, where value is the lesser of either appraised value or sales price.
 - $\text{LTVR} = (\text{loan amount} \div \text{value}) \times 100$

Example

A buyer borrows \$315,000 to purchase a \$350,000 property.

$$\text{LTVR} = \$315,000 \div \$350,000 = .9 \text{ or } 90\%$$

- $\text{Loan amount} = \text{LTVR} \times \text{value}$

Example

Property value is \$350,000.

LTVR is 90%.

$$\text{Loan} = \$350,000 \times 90\% = \$315,000$$

- $\text{Value} = \text{loan amount} \div \text{LTVR}$

Example

Buyer borrows \$315,000. LTVR is 90%. $\text{Value} = \$315,000 \div 90\% = \$350,000$.

10. Down Payment/Amount to Be Financed

- $\text{Amount of down payment} + \text{earnest money deposit} + \text{loan amount being assumed or obtained} = \text{sale price}$.
- The down payment is calculated as a percentage of the sales price based on the lender-required LTV.
 - $\text{Sales price} - \text{down payment} - \text{earnest money deposit} = \text{amount to be financed}$

11. Discount Points and Loan Origination Fees

- Discount points are pre-paid interest paid by or on behalf of borrowers at the beginning of the loan to reduce monthly mortgage payments.
 - Discount points (pre-paid interest) = 1% of the loan amount per point
- To lower the interest rate, borrowers can purchase discount points. Each discount point costs 1% of the loan amount and is a discount of 0.25% off the mortgage rate (this is referred to as the rule of thumb in the lending world). For example, to lower the mortgage interest rate from 5% to 4.25%, the borrower would need to pay three discount points, which would then lower the mortgage interest rate 0.75%.
 - Amount paid for points = loan amount $\times$ number of points (followed by a % sign)

Example

A buyer purchases a \$350,000 property and gets a \$315,000 loan.

Two discount points is $2\% \times \$315,000 = \$6,300$

For additional information, consult your PSI candidate information bulletin.